TOYING WITH RELATIONS: SPECULATIVE TOYS FOR CHILDREN AND SUSTAINABILITY

KAREY HELMS
STOCKHOLM UNIVERSITY, UMEÅ UNIVERSITY
KAREY.HELMS@UMU.SE

PETRA JÄÄSKELÄINEN
KTH ROYAL INSTITUTE OF TECHNOLOGY
PPJA@KTH.SE

SHARON LINDBERG
STOCKHOLM UNIVERSITY
SHARON@DSV.SU.SE

ELIN SPORRONG
STOCKHOLM UNIVERSITY
ELIN.SPORRONG@DSV.SU.SE

ARJUN MENON
KTH ROYAL INSTITUTE OF TECHNOLOGY
ARMENON@KTH.SE

MEIKE SCHALK
KTH ROYAL INSTITUTE OF TECHNOLOGY
MEIKE.SCHALK@ARCH.KTH.SE

AIRI LAMPINEN
STOCKHOLM UNIVERSITY
AIRI@DSV.SU.SE

ABSTRACT

Toys are situated within the relational ontologies of children, which includes how more-than-human relations are imagined and valued through play. Motivated by a concern for a damaged environment, we critique a selection of existing toys intended for outdoor play and fabulate other versions based upon alternative relations. From the detailing of six redesigns through text and collage, we foreground generative pathways for how children, and adults, might relate differently to nature. These pathways are presented as questions prompted by the toys. This exploratory research contributes to the imagining of multispecies worlds and the agency of children in toying with relations.

INTRODUCTION

"Toy" is a noun and a verb: it refers both to an object and a way of relating (Woodall, 2022). As objects, toys are for many beings and take many forms. From simple shapes to industrialized objects to interactive devices, they are things to play with. Within the relational ontologies of children, play is something to take seriously (Vygotsky, 1967; Bento & Dias, 2017) and toys are a significant site for engaging with posthuman values and concerns (Hickey-Moody et al., 2021). This means that toys and their implications are not bounded or discrete within playtime. Instead, toys are socially, politically, and materially entangled in their impacts on children and the world.

Our interest in toys is as a pathway to explore more-than-human relations, and particularly how children might engage differently with the outdoors and its multispecies inhabitants. We view children as citizens, dwellers, and caretakers of the earth (Hickey-Moody et al., 2021). This dual attention to entanglements with nature and the agency of children situates a posthuman design space to toy with relations.

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In building upon design research for children and environmental sustainability (Edwards et al., 2021; Cumbo & Iversen, 2020; Walker, 2017), we critique a selection of existing toys intended for outdoor play to propose other versions based upon alternative human-nature relations. We ask: What are the values and relations foregrounded in a selection of existing toys? How might opposing values and alternative relations ground other versions of these toys? And what pathways in designing for posthuman children and sustainability would these versions open for?

In the following, we explore these questions through the design of six speculative toys. Each is based upon a critique of an existing toy and a proposed other version through a visual collage and written description. The redesigns are fabulations (Søndergaard et al., 2023) that imagine how outdoor play might be otherwise. From them, we reflect upon the speculative toys as presenting children with opportunities to affect more-than-human relations and ask questions through play.

This exploratory paper is situated among multispecies world-making practices (Westerlaken, 2020), fabulating more-than-human entanglements (Helms et al., 2021), and the making of posthuman theory through design (Nicenboim et al., 2023; Helms, 2024). It offers six speculative toys as instantiations of a design space for posthuman children and sustainability, and it contributes the questions that the toys pose as possible generative pathways to design for how children, and adults, might relate to nature differently.

SIX SPECULATIVE TOYS FOR CHILDREN

Our research-through-design process is situated within interaction design has included 1) the systematic selecting of existing toys for outdoor play, 2) multiple workshops with design researchers in which these toys were critiqued and initial redesigns proposed, and 3) an analysis of workshop data and subsequent redesigns. The resulting toys are part of an ongoing process investigating new definitions of digital experiences in the outdoors and with a planned future inclusion of children. Electrical dependency might appear at odds with environmental sustainability, yet we motivate its inclusion alongside calls to investigate digital technology in early childhood (Vella et al., 2023).

We conceptualize our process as a design space in which artifacts (i.e. toys) are organized, reprogrammed, and transformed through conceptual and material activities (Binder et al., 2011). In this exploratory paper, we do not detail our process further as our intention is not to validate the proposed redesigns as “better” toys than the originals. Instead, we first highlight two methodological commitments, critique and fabulation, and then describe how each speculative toy will be presented.

We draw upon critique as an intentional search for value (Bardzell & Bardzell, 2015). In our critique of existing toys, we employed interaction criticism to investigate the complex relationship between the aesthetics of interaction, material meanings, and social considerations (Bardzell, 2011). More concretely, we identified assumptions and functionalities, and then considered opposites of these to ground the speculating of other versions. Fabulations imagine alternative ways of living and representing pasts and presents (Rosner, 2018), and have a rich history of speculating upon and within more-than-human relations (Despret, 2016; Haraway, 2016). We draw upon aims to stimulate imagination through joy amid serious concerns (Søndergaard et al., 2023), with an openness towards unknown possibilities in the form of lingering questions (Helms et al., 2021).

For each proposed redesign, we first position the new toy as a different version of an existing toy. We then briefly describe the existing toy and a critique of it. While some existing toys might be considered tools, we do not attend to this difference. Instead, we stay close to their potential for play as an exploration of relating. Next, we describe the new toy, which includes highlighting key elements in its accompanying collage and portraying possible skills, benefits, and values of the speculative version. As fabulations, this can be seen as idealist in the imagining of how we as adults would like people to be entangled in multispecies play.

LEAF-STITCH KIT

Leaf-stitch kit (Figure 1) is a different version of a pocketknife. A pocketknife is a foldable multi-tool with a saw, knife, scissors, and awl to physically cut or shape materials found in nature. Opposite functions of a



Figure 1: *Leaf-stitch kit* is another version of a pocket knife. It is a foldable multi-tool with scissors, thread, and needles that can be used for repairing and mending leaves.

pocketknife are reattachment, reassembly, or regeneration.

The leaf-stitch kit is a foldable multi-tool with scissors, thread, and needles to mend, heal, and repair plants. A scenario of play is children in the forest stitching torn leaves back together. In the collage, children are shown collaboratively stitching together leaves and the bottom background image depicts what this might look like. An image of a cow eating leaves in the top left prompts as to why leaves might be in certain conditions. The details in top right background image, while actually a zoomed-in photograph of a fly, are intended to represent intricate, delicate, and decorative stitching and embroidery that might evolve with practice.

Possible skills, benefits, and values of the leaf-stitch kit include cultivating nurture and discovering traces of nonhuman activity, such as animals, caterpillars, or diseases on trees. The kit also toys with the "myth of fixing" whereby maintenance and imperfection in nature becomes something magical and delicate.

DARKNESS MASK

Darkness mask (Figure 2) is another version of a snorkel. The snorkel is a mask that looks like a shark with an orange mouth tube. It is for being close to unfamiliar things underwater. An opposite is assuming children are powerful and that distances between species should be maintained while exploring another's environment.



Figure 2: *Darkness mask* is another version of a snorkel. The mask augments senses of hearing and smelling in unfamiliar and vulnerable atmospheres, such as the dark.

The darkness mask augments nighttime sounds and scents based on what species might live in the habitat and be active. Like the ocean, the dark is an unfamiliar and vulnerable atmosphere. A scenario of play is a child pretending to be an owl while camping in the forest. The child is heroine of the night and feels powerful. In the collage, the background images convey a spooky space of shadows and indistinguishable figures, and the stars in the top left hint at even bigger and further unknown atmospheres. The child stands proud. An owl, a badger, and a spider are nocturnal companions.

Possible skills, benefits, and values of the darkness mask are exploring bodily ways of being in different atmospheres through the defamiliarization of senses. Being in the dark exposes the child to routines and ways of living outside standardized human clock-time.

FERAL FAN

Feral fan (Figure 3) is a redesign of gardening tools. Gardening tools are a shovel, spade, and rake that can be worn in a belt. They promote learning about cycles of growth, nutrition and optimal placement of plants. Opposites of this are going feral and making wild.



Figure 3: *Feral fan* is another version of gardening tools. It uses the wind to making otherwise domesticated plants feral and promotes unpredictable flourishing.

The feral fan is for a child to pour seeds into, which are indeterminately scattered in the wind. A scenario of play is a child running around a field with the fan. It shifts the emphasis of gardening away from digging, planting, weeding, and watering, to using airflow to making domesticated plants feral. In the collage, the background images depict a mix of flowers and weeds as a child runs and scatters seeds with the help of the wind. The increase of plants and insects from one side of the collage to the other hints at gradual growth, unpredictable flourishing, and uncertainty regarding how far seeds may travel.

Possible skills, benefits, and values of the feral fan are the inclusion of the wind as an invisible, yet felt, participant. It encourages movement, indeterminacy, and waiting. It is about welcoming rather than weeding, or inclusion rather than exclusion.

HUMAN CALL

The human call (Figure 4) is a redesign of a bird call. The bird call makes realistic bird sounds by twisting a metal knob. It promotes an audible sensitivity and attentiveness in learning about different sounds. An opposite interaction is a human sound-making device for birds.

The human call is for children to put in trees, like bird houses, for birds to call children. A scenario of play is children hearing sounds from the human call and running to the tree in response. In the collage, a bird is shown operating the crank in the top right, two bird bystanders are in the top left, and various children running and climbing trees in response. The background images depict forests and parks to convey places that children might be near or within. The collage conveys urgency and enthusiasm with which children respond, and various birds assembling to watch the children.



Figure 4: *Human call* is another version of a bird call. It is a toy for birds that children can put up in trees, like bird houses, for birds to call for children.

Possible skills, benefits, and values of the human call is its orientation as a toy for children to give as a toy to birds. While it assumes that birds will care about it – or be able and willing to use it – the human call celebrates interspecies communication as playful and is grounded in the imagination of children.

SNAIL TRAIL LIGHT

Snail trail light (Figure 5) is another version of chalk. Chalk are colored sticks meant for drawing on hard surfaces outside. It assumes the outdoors is a passive and neutral place upon which one can impose ideas and decoration. An opposite is discovering invisible art created by nonhumans that is purposeful for a particular being.



Figure 5: *Snail trail light* is another version of chalk. It is a UV flash light for discovering “art” in nature, such as snail trails and animal pee.

The snail trail light is a UV flashlight for discovering art, such as snail trails and animal pee. A scenario of play is a child investigating the everyday traces of nonhumans moving, urinating, or releasing bodily substances as part of their being in the world. In the collage, a child is shown shining a light to illuminate slime trails from snails and slugs. The curvy patterns of the trails are appreciated and make visible the liveliness of beings that people often consider slow. The background images depict leaves and rocks as surfaces on which trails might be. These prompt the idea that any outdoor surface has the capacity to reveal something.

Possible skills, benefits, and values of the snail trail light are discovering things that are often considered disgusting and highlighting them as worthy of appreciation. It promotes paying attention to surroundings for aesthetic reasons and curiosity.

SOMEWHERE SOUND KALEIDOSCOPE

SomewHere sound kaleidoscope (Figure 6) is another version of an insect jar. The insect jar is a container to put insects in and look at through a microscope. It encourages learning about small living things while promoting human mastery over nature. An opposite function of the insect jar is a child “being” in the world of nonhuman.

The somewHere sound kaleidoscope is a listening device. It magnifies a sound from a child’s immediate environment, which might include their own breathing or squiggling of toes, and a magnified sound from someplace else, such as the blinking of a lizard. The sound from someplace else is chosen by turning a knob that controls the distance from the child’s location to an unknown place. The toy randomly picks up a sound from that unidentified location to magnify and mix audio, acting as an audible (rather than optical) kaleidoscope. In the collage, a child is shown laying on

the ground, somewhere outside, while holding the toy to their ear. The background images are zoomed in images of a fish, lizard, and grasshopper as what the child might be listening to.



Figure 6: *SomewHere sound kaleidoscope* is another version of an inset jar. It is for a child to listen to a sound from nearby and a sound from someplace else.

Possible skills, benefits, and values of the somewHere sound kaleidoscope are the challenging of outside as the same outside everywhere, and not knowing what a child might find. Like a radio dial, tuning the kaleidoscope into different places has a creative element of mixing, which brings up tensions between different sounds or the seeking of harmony between a there and here.

WHAT WOULD CHILDREN PLAY IF GIVEN THE RIGHT TOY?

We have presented six speculative toys for children that foreground relations with nonhumans, outdoor spaces, and natural materials. Although each fabulation is based upon a critique of an existing toy, we do not advocate for them as "better" versions. In drawing upon *What would animals say if we asked the right questions?* (Despret, 2016), the redesigns are intended as *additive*: they are other versions that present children with opportunities to ask different questions about their relations with nature and multispecies worlds. As Taylor writes, contextualizing these ideas within design research, and which we apply to the interests of children: "*Design might offer up the space not to enforce (from the outside) what should interest [children], but to begin wondering how we might be open to the possibilities of what could become interesting, of how to develop, twist and cultivate these possibilities between an artefact and its user*" (Taylor, 2017).

While Despret focuses on the interests and capabilities of animals, the toys presented here as fabulations remain toys for children. Through them, we formulate the provocation *What would children play if given the right toy?* This framing considers toys as presenting children with opportunities to ask different questions and acknowledges the agency of children in affecting relations regardless of the "right" toy. The speculative versions stay with design research as a negotiating of possibilities and an asking of questions, probing into how the possibilities of different questions might cultivate other ways of being and imagining multispecies worlds (Westerlaken, 2020). In other words, how might we design toys that give children the chance to playfully ask different questions to animals, plants, adults, themselves, or each other?

For each of the six toys, we now highlight our own questions stemming from them as possible generative pathways for designers. The questions are speculations about the interests of children, and how design might be inspired by, or participate in bringing forth, such speculations.

GENERATIVE PATHWAYS FOR POSTHUMAN CHILDREN AND SUSTAINABILITY

Leaf-stitch kit prompted us to ask: What does a child want to put back together? Why should a leaf need this? How is repair relevant for such things, if at all? From these, we reflected upon human cycles of repair, like pruning an apple tree, and how we repair what we see or need, finding difficulty grasping other regenerative timescales, such as volcano cycles. This points to a superfluousness in the toy as a delicate form of mending or fixing, yet it also proposes that perhaps environmental care does not need to be pragmatic or utilitarian.

Darkness mask prompted us to ask: Whose environment is this? Are they disturbed by a child being in it? Who is kept safe, the child or nocturnal beings? From these, we reflected upon encountering ways of living and liveliness that are different than a Western child's world of routines, such as encountering the dark when autumn arrives in Northern Europe, or unexpectedly on the way home from a trip. We discussed different forms of dark, what the absence of light enables, who inhabits and thrives in these conditions, and how it might be possible to cultivate multispecies respect while exploring unknown environments.

Feral fan prompted us to ask: What grows together? Why is knowledge about this needed? And what is the difference between invasion and inclusion? From these, we reflected upon relinquishing control through not knowing where things will grow, celebrating movement and flow, and waiting while hoping that things will flourish. This foregrounds material, social, and political belonging and thriving in an environment: to be exotic,

native, both, or neither, in making space for oneself. We considered how invisible forces, such as the wind, collaborate with the feral fan and cultivate play.

Human call prompted us to ask: How does it sound to hear a language you do not understand? Do all humans sound the same to a bird? What would birds assume children want, if they assume at all? From these, we reflected upon stereotypes in the categorization of languages, the background "noise" of indistinguishable human mumbling, and the presumption that birds are curious enough about us to put in the effort of calling. Changing our minds, we also asked: Why not imagine birds as cunning and playful? Perhaps children and birds are conspiring to summon adults by indistinguishable human mumbling?

Snail trail light prompted us to ask: What if we moved from instructions to spaces for play? And, as mentioned above, what is found or created in emptiness? From these, we reflected upon invisible traces and noticing through making, such as in the project "Design by Animals" (Front, n.d.) in which vases were created from castings of animal burrows and holes. We build upon Ávila (2022) in considering the snail trail light as a form of co-fabrication in an ethics of noticing and relating.

SomewHere sound kaleidoscope prompted us to ask: What, or where, does a child want to be? How does it feel to be oneself and something else? Can places, plants, animals, or ourselves be captured through senses other than seeing? From these, we reflected upon collecting seashells to listen to the ocean, as portals to other worlds, while trying to find harmony between being there and somewhere else. This points towards the imagining of being otherwise, without the assumption of that is what it is like to be something or somewhere else.

CONCLUSION

We have presented six speculative toys in the form of text and collage to fabulate how human-nature relations might be otherwise. We did this through a critique of existing toys for children intended for outdoor play, and from which we brought forth alternative values and relations to ground the design of other versions. Each redesign is a concrete opportunity for how children, and adults, might relate to nature differently and contributes questions as generative pathways forward in designing for more-than-human relations. This work is situated within interaction design research for children and sustainability, and contributes to the agency of children in toying with multispecies worlds.

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